

GENERAL HYDRAULIC NOTES

G1	ALL HYDRAULIC SERVICES SHALL BE CARRIED OUT IN ACCORDANCE WITH , NATIONAL CONSTRUCTION CODE, APPROVAL OF THE SITE SUPERINTENDENT, AS3500.1,2,4 & 5.
G2	THIS PLAN SHALL BE READ IN CONJUNCTION WITH APPROVED ARCHITECTURAL, STRUCTURAL, CIVIL, ELECTRICAL MECHANICAL, LANDSCAPE DRAWINGS AND SPECIFICATIONS.
G3	ALL SANITARY DRAINAGE LINES ARE TO BE LAID ON A MINIMUM OF 100MM CLEAN SAND BEDDING ATOP OF A SOLID GROUND BASE. IF OVER EXCAVATION IS EXPERIENCED COMPACTED FILL TO BASE LEVEL WILL BE REQUIRED.
G4	ALL SANITARY DRAINAGE AND SANITARY PLUMBING LINES SHALL BE SUBJECT TO A WATER STATIC TEST FOR 1 HOUR AS PER AS35002. PROVIDE A WITNESS MARK TO THE LEVEL OF THE WATER AT COMMENCEMENT OF THE TEST. WATER TEST IS TO REMAIN IN PIPELINES UNTIL TRENCH REINSTATEMENT IS COMPLETE. ON COMPLETION PROVIDE A DRAINAGE TEST CERTIFICATE AS PART OF THESE WORKS ACCOMPANIED BY PHOTOGRAPHIC EVIDENCE SIGNED OFF BY A THIRD PARTY.
G5	PROVIDE SANITARY DRAINAGE INSPECTION OPENINGS TO FINISHED SURFACE LEVEL COMPLETE WITH APPROVED ACCESS COVERS.
G6	THE LOCATION OF EXISTING WATER AND SANITARY SERVICES SHOWN ARE APPROXIMATE ONLY AND SHALL BE VERIFIED ON SITE PRIOR TO COMMENCEMENT OF WORKS. REDUNDANT EXISTING SERVICES SHALL BE DISCONNECTED AND CAPPED AT THE LIVE MAIN AND REMOVED WHERE FEASIBLE. IT IS THE RESPONSIBILITY OF THE PLUMBING CONTRACTOR TO APPRAISE THE SITE TO DETERMINE THE DRAINAGE CONFIGURATION AND ALLOW MAKING ADJUSTMENTS TO SUIT.
G7	ALL SANITARY DRAINAGE LINES ARE BE SET-OUT IN ACCORDANCE WITH AS3500.2 IN REFERENCE TO INSTALLING DRAINS NEAR OR UNDER BUILDINGS.
G8	ALL SERVICES THAT CROSS PAVEMENTS,FOOTINGS ETC. SHALL BE BACKFILLED WITH GRANULAR MATERIAL TO SUBGRADE LEVEL AND COMPACTED TO 98% M.M.D.D.
G9	ON COMPLETION OF PIPE INSTALLATION, ALL DISTURBED AREAS MUST BE RESTORED TO ORIGINAL CONDITION INCLUDING KERBS, FOOTPATHS, CONCRETE AREAS, GRAVEL AREAS, GRASSED AREAS AND ROAD PAVEMENTS.
G10	TRENCHES THROUGH EXISTING ROAD AND CONCRETE AREAS SHALL BE SAW-CUT TO FULL DEPTH OF CONCRETE AND A MIN. OF 50mm IN BITUMINOUS PAVING. REINSTATE WITH ADDITIONAL REINFORCEMENT & DOWLING AS REQUIRED BY STRUCTURAL ENGINEERS. FINISH TO MATCH EXISTING. REINSTATE SURFACE COVERINGS NOT OTHERWISE TO BE REPLACED.
G11	CARE SHALL BE TAKEN WHEN EXCAVATING NEAR EXISTING SERVICES. NO MECHANICAL EXCAVATION SHALL BE TAKEN OVER TELSTRA OR ELECTRICAL SERVICES. EXCAVATE BY HAND IN THESE AREAS.
G12	ALL uPVC PIPE EXPOSED TO SUNLIGHT SHALL BE DWV SN6 AS A MINIMUM.
G13	CONTRACTOR IS RESPONSIBLE FOR ANY RELEVANT AUTHORITY APPROVALS AND PAYMENT OF ALL ASSOCIATED FEES.
G14	THE CONTRACTOR SHALL ALLOW FOR ALL TIMBERING, SHORING AND SHUTTERING AS NECESSARY TO CONSTRUCT PIPEWORK INCLUDING THE REMOVAL OF SAME UPON COMPLETION OF PIPEWORK.
G15	PROVIDE 80mm COMPRESSIBLE MATERIAL UNDER CONCRETE PATHWAYS AND FOUNDATION WHERE PIPEWORK CLEARANCE TO UNDERSIDE OF FOOTING IS LESS THAN 150mm (AND GREATER THAN 75mm).
G16	PRIOR TO PRACTICAL COMPLETION OF WORK, THE CONTRACTOR IS TO SUPPLY A FULL SET OF "AS CONSTRUCTED" DRAWINGS COMPLETE WITH ACCURATE LOCATION AND DEPTHS OF PIPELINES.
G17	ANY PENETRATION THROUGH FIRE RATED WALLS/FLOORS/CEILINGS ARE TO BE FIRE RATED TO EQUAL RATING.
G18	ANY PENETRATION THROUGH WALLS OR FLOORS TO BE SEALED WITH A SOLID MATERIAL (NOT SILICONE) TO PREVENT ENTRY OF INSECTS & VERMIN.

SANITARY DRAINAGE NOTES

SD1	CONTRACTOR SHALL ALLOW FOR ADDITIONAL EXCAVATION, BACKFILL OF PIPES, FITTINGS AND ALL JUMP UPS AS REQUIRED BY THE NATIONAL PLUMBING AND DRAINAGE CODE AS3500.2.
SD2	ALL SANITARY DRAINAGE PIPEWORK SHALL BE Ø100mm uPVC SN6 AT 1.65% MIN. GRADE UNLESS NOTED OTHERWISE.
SD3	ALL BRANCH LINES SHALL BE GRADED AND/OR LOWERED TO AVOID PENETRATING FOOTING BEAMS UNLESS SHOWN OTHERWISE. THE CONTRACTOR SHALL ALLOW FOR ADDITIONAL PIPE LENGTHS AND WORK INCLUDING CONSTRUCTION OF ADDITIONAL INSPECTION OPENINGS AS REQUIRED BY AS3500.2.
SD4	ALL SANITARY DRAINAGE INSTALLATIONS TO BE COORDINATED WITH THE STRUCTURAL DRAWINGS. NO STEEL REINFORCING BARS TO BE CUT WITHOUT PRIOR WRITTEN APPROVAL FROM STRUCTURAL ENGINEER.
SD5	ALL DRAINS INSTALLED IN GROUND THAT IS FILLED, UNSTABLE, EXPANSIVE OR WATER CHARGED AND WHERE SOIL MOVEMENT MAY AFFECT THE PERFORMANCE SHALL BE INSTALLED TO BDAQ "RECOMMENDED GUIDELINES FOR SANITARY & STORMWATER INSTALLATION ON REACTIVE CLAY SITES".
SD7	ALL INSPECTION OPENINGS ARE TO BE INSTALLED TO SUIT THE INSTALLATION SURROUNDING IN ACCORDANCE WITH AS3500.2. BOLTED TRAP SCREW INSPECTION OPENINGS COMPLETE WITH BRASS LIDS ARE TO BE USED IN THE MOWING STRIP LOCATIONS AS SHOWN ON THE DESIGN DRAWINGS. FINISHED FLUSH WITH SURROUNDING SURFACE.
SD8	SANITARY DRAINAGE VENT PIPES TO BE RUN INTERNALLY WITHIN WALLS EXTENDING TO ATMOSPHERE WITH MATCHING ROOF COLOUR PENETRATION FLASHINGS. ALL WORKS IN ACCORDANCE TO AS3500.2.

WATER SERVICES NOTES

WS1	REMOVE EXISTING REDUNDANT SERVICE PIPEWORK AND REDIRECT EXISTING BURIED WATER SERVICE TO NEW SERVICES GENERALLY AS SHOWN.
WS2	ABOVE GROUND, INTERNAL DRINKING HOT & COLD WATER SYSTEM <= DN22 SHALL BE PE-X PN16 & ARE NOMINATED ON DRAWINGS AS OUTSIDE DIAMETER. INSTALL PE-X PLUMBING SYSTEM IN ACCORDANCE WITH THE PE-X PLUMBERS TECHNICAL & INSTALLATION MANUAL. PIPES > DN22 TO BE 316 STAINLESS STEEL WITH PRESS FIT COUPLINGS.
WS3	ALL EXTERNAL EXPOSED COLD WATER SERVICE CONNECTION PIPES ABOVE GROUND ARE TO BE GRADE 316 STAINLESS STEEL CONNECTORS AND UNIONS.
WS4	ALL EXTERNAL EXPOSED HOT WATER SERVICE CONNECTION PIPES ABOVE GROUND ARE TO BE GRADE 316 STAINLESS STEEL CONNECTORS AND UNIONS.
WS5	INSULATION: ALL HOT WATER PIPES SHALL BE INSTALLED IN ACCORDANCE WITH THE NATIONAL CONSTRUCTION CODE OF AUSTRALIA
WS6	ALL EXTERNAL HOSE TAPS ARE TO BE INSTALLED WITH GE ZB007 VACUUM BREAKERS IN ACCORDANCE WITH AS 3500.1.
WS7	LOCATE HOSE TAPS AS SHOWN ON THE DRAWING.
WS8	FOR SLAB ON GROUND AREAS, ALL WATER SERVICES TO BE RETICULATED THROUGH CEILING SPACE OR UNDER ROOF AND DROPPED FROM ABOVE. WATER SERVICES MAY BE RUN UNDER SUSPENDED FLOORS WITH RISERS IN WALLS. NO LONGITUDINAL SERVICES THROUGH WALLS WILL BE ACCEPTABLE.
WS9	ALL INTERNAL HOT & COLD WATER RISERS & DROPPERS (INSIDE BLOCK WORK) TO BE INSTALLED IN APPROPRIATELY SIZED CONDUIT WITH LONG RADIUS BENDS OR DUCTS FOR EASE OF REPLACEMENT.
WS10	INSTALL HARSMITH BUILDING PRODUCTS 'FLUID APRON' (OR EQUAL) TO ALL WALL MOUNTED TAPS & WALL ELBOWS.

INSTALLATION GUIDELINES FOR DRAINAGE SYSTEM INSTALLED IN REACTIVE SOIL

FOR DRAINAGE SYSTEM INSTALLED IN REACTIVE SOILS, "STORM PLASTICS SWIVEL EXPANSION AND EXPANSION JOINTS" OR EQUAL APPROVED TYPE FLEXIBLE JOINTS SHALL BE INSTALLED.

WHERE DRAINAGE ARE SUPPORTED AND BRACKETED TO UNDERSIDE OF BUILDING FRAMES, MECHANICAL FLEXIBLE JOINTS SHALL BE PROVIDED.

WHERE DRAINS ARE INSTALLED UNDERSIDE OF CONCRETE SLAB, COMBO JOINTS FOR RISER AND EXPANDA JOINTS ON HORIZONTAL DRAINS SHALL BE FITTED IN ACCORDANCE WITH AS/NZS 2032-2006 AND MANUFACTURE'S RECOMMENDATIONS.

WHERE DRAINS ARE INSTALLED OUTSIDE OF THE BUILDING, EXPANDA/SWIVEL JOINTS SHALL BE PROVIDED IN ACCORDANCE WITH AS/NZS 2032-2006 AND MANUFACTURES'S RECOMMENDATIONS. ENSURE ADEQUATE LAGGING TO JOINTS IS PROVIDED.

FOR DOWNPIPES CONTINUED AND CONNECTED TO IN GROUND STORM WATER PIPES, PROVISION SHALL BE MADE FOR EXPANSION AND CONTRACTION AT SPACES NOT EXCEEDING 6 METERS.

FOR TENDER

ALL DIMENSIONS SHOWN IN MM. DO NOT SCALE FROM DRAWINGS. DIMENSIONS TAKE PRECEDENCE. IF IN DOUBT REFER TO HK SOLUTIONS.

T1	ISSUE FOR TENDER	17.06.2026	MC	HKS	ENGINEERING PROJECT MANAGEMENT DARWIN,GEELONG, CAIRNS,TOWNSVILLE, PERTH,BALLARAT admin@hksolutions.com.au www.hksolutions.com.au Conditions of use. This document may only be used by Harris Kmon Solutions clients for the purpose for which it was prepared and must not be used by any other person or for any other purpose.
REV.	DESCRIPTION	DATE	INT.	COMPANY	
REVISIONS					

SPORTENG™

DRAWN	AL
CHECKED	MC
DESIGNED	MC
CHECKED	LM
APPROVED	

THIS DRAWING MUST NOT BE USED FOR CONSTRUCTION UNLESS SIGNED AS APPROVED

STAGE 1
ROCKHAMPTON SPORTS PRECINCT - NETBALL CLUBHOUSE
554-700 YAAMBA ROAD, NORMAN GARDEN - 4701
HYDRAULIC NOTES

PAGE No	PROJECT No - DRAWING No	DISCIPLINE	REV.	SHEET SIZE
03 of 10	25015-H-NBC-002	HYDRAULIC	T1	A1

DRAWINGS TO BE PRINTED IN COLOUR